

P r o b l e m S e t # 2

1. iPad Pricing & Advertising

The annual demand for Apple iPads is given by $Q = 2000(1000 - P)$. The marginal cost to Apple of making and selling an iPad is \$200.

- a) What price should Apple set for the iPad to maximize profits? At this price, how much will iPad sales contribute towards Apple's R&D costs and other fixed costs?
- b) Does this price depend upon the costs Apple incurred to develop the iPad? Does it depend on the cost that Apple expects to incur to develop the next generation iPad?

The marketing team at Apple is considering running an advertising campaign with the catch phrase "Join the iPad World." The campaign would be designed to expand the target market for iPad customers. The campaign would cause the demand for iPads to expand by 10%. More precisely, after the campaign the demand for Apple iPads would be $Q = 2200(1000 - P)$.

- c) If Apple does run the "Join the iPad World" campaign, should they charge a different price for the iPad than in part (a)? If so, calculate the new price. If not, explain why not.
- d) What is the most that Apple would pay to run the "Join the iPad World" campaign?

An alternative advertising campaign would use the catch phrase "I Love My iPad." This campaign would target traditional iPad customers, emphasizing the wonderful things that iPads can do. This campaign would increase the willingness to pay of iPad customers by 10%. More precisely, after this campaign the demand for Apple iPads would be $Q = 2000(1100 - P)$.

- e) If Apple does run the "I Love My iPad" campaign, should they charge a different price for the iPad than in part (a)? If so, calculate the new price. If not, explain why not.
- f) What is the most that Apple would pay to run the "I Love My iPad" campaign?

2. Tesla

Tesla Motors designs, develops, manufactures, and sells high-performance, fully electric vehicles. In the six months ending June 30, 2013, Tesla recognized \$957 million of revenues from automotive sales, based primarily on delivery of 10,050 of its Model S vehicles. Tesla reported the cost of these revenues as \$765 million, leaving a gross profit on automotive sales of \$192 million for this six-month period. Gross profit does not include R&D expenses or selling, general and administrative expenses. Including those expenses, Tesla took a net loss during the first half of 2013 of \$19 million.

Tesla benefits from a program operated by the California Air Resources Board requiring that 15% of vehicles have zero emissions by 2025 – up from about 1% currently. (Other states have similar programs.) Manufacturers that fall short of that goal can avoid paying penalties by purchasing “Zero Emission Vehicle” (ZEV) credits. Since Tesla only makes ZEVs, Tesla has credits to sell. Included in Tesla’s \$957 million of reported revenue from “automotive sales” during the first half of 2013 was \$119 million that Tesla received by selling such regulatory credits to other automobile manufacturers, including Honda. Accordingly to some reports, Tesla receives \$35,000 to \$45,000 per vehicle for the sale of regulatory credits, but this figure has not been confirmed by Tesla.

- a) Should the price Tesla receives for ZEV credits affect the price Tesla charges for its Model S vehicles? If so, explain why. If not, explain why not.

Separately, Tesla had previously received a loan from the Department of Energy for \$465 million under the Advanced Technology Vehicles Manufacturing Incentive Program. In May 2013, Tesla entirely paid off this outstanding loan, replacing it with new financing.

- b) Assume that Tesla’s new financing is on more favorable terms than the DOE loan, thus reducing Tesla’s monthly interest expenses. Should this affect the price Tesla charges for its Model S vehicles? If so, explain why. If not, explain why not.
- c) Suppose now that Tesla is delivering as many vehicles as it can make given its current production capacity. Would that change your answer to part (a) and/or part (b)?

3. Focus

PharmCo has developed a new drug, Focus® that increases energy levels, alertness, and problem solving abilities for up to four hours. The only known side effect is exhaustion for the subsequent 12 hours. Sadly, Focus® is only safe for those between 18 and 30 years of age. PharmCo estimates that the marginal cost of producing and selling one dose of Focus® is \$1.

The military is very interested in purchasing Focus®. A much larger student market also looks very promising. Government procurement rules prohibit the military from paying more for Focus® than civilians pay. Assume PharmCo is thus forced to set a single price for Focus®.

In the first year following approval by the Food and Drug Administration (FDA), PharmCo set the price of Focus® at \$25 per dose, selling 1 million doses to the military and 9 million doses to students. Based on market surveys and some limited pricing experiments, PharmCo estimates that the absolute value of the elasticity of demand for Focus® by the military at a price of \$25 per dose is 0.2, and that the absolute value of the elasticity of demand for Focus® by the students at a price of \$25 per dose is 1.2.

- a) Would PharmCo’s profits go up or down if it raised the price of Focus® slightly above \$25 per dose? Explain, showing your calculations.
- b) Some people at PharmCo are concerned about future product liability lawsuits associated with the sales of Focus®. You have seen one estimate – a guess really – that for every million doses sold, PharmCo will incur expected liability of \$2 million. Using that figure, would your answer to part (a) change? Explain, showing your calculations.
- c) Would risk aversion by PharmCo affect your answer to part (b)?

4. California Memorial Stadium

The California Memorial Stadium has a capacity of 60,000 and is used for exactly seven football games per year. For simplicity, assume there is single price for all seats at a given game, but prices can vary from game to game. Three of the games each season are Pac 12 Games. The demand for tickets for each Pac12 Game is given by $Q = 150,000 - 3000P$ where P is ticket price. Three of the games each season are Non-Conference Games. The demand for tickets for each Non-Conference Game is given by $Q = 90,000 - 3000P$. There is also one Big Game each year. The demand for tickets at the Big Game is given by $Q = 240,000 - 3000P$. Assume that the cost of operating the stadium is independent of the number of tickets sold. (Equivalently, the marginal cost of having an additional spectator is just offset by the profit margin the stadium makes on the food and drinks that person purchases while at the game.) Assume ticket prices are set to maximize profits, which are used to support football and other sports programs at Cal.

- a) What are the profit-maximizing prices for the three types of games?
- b) Since the Stadium is frequently full, you would like to squeeze in more seats. Suppose you can add one new seat to the stadium, just for this year, for \$75. Would this raise profits? Hint: Think about the marginal revenue from the extra seat.

5. Price-Bots

You can sell into two separate markets with no arbitrage or resale between markets. An individual in Market A has the following annual demand: $q = 50 - p$. An individual in Market B has the following annual demand: $q = 120 - 2p$. Market A has 4000 people in it. Market B has 1200 people in it. You are the only producer. Total cost of producing q units is $TC = 5000 + 20q$

- (a) How many units do sell in A and at what price?
- (b) How many units do sell in B and at what price? What are your profits in A and B.
- (c) Thanks to the internet, the wall between markets has fallen. Consumers in both A and B have unleashed an army of 'price-bots' so they will be able to instantly know about and be able to purchase your good at the lowest price you offer in either market. What price do you charge in each market and how much do you sell? What happens to your profits? Why?

6. Golf

You are asked to help with the pricing at a golf course. The course has 10,000 customers, each with an individual monthly demand curve for games played $q(p) = 10 - p$. The marginal cost for a game played is \$2, and monthly overhead costs are \$150,000. The course has enough capacity to serve all potential customers without overcrowding the course. Assume that the fixed cost for serving each customer (such as the cost of printing a membership card) is negligible, so that all variable costs are due to games played.

- (a) The owner of the course is an old-fashioned businessperson and insists on a simple pricing strategy, i.e., a fixed price per game played. Your job is just to figure out the optimal price. What price would you set, and what would be the monthly profits?
- (b) Emboldened by your success in part (a), you decide to show the owner that the course would be better off by implementing a membership fee and a price per game. What would be the optimal prices, and what would be the monthly profits? Also, try to briefly explain to the owner the intuition of why this works better than simple pricing.

7. Carrot Juice

You own the Golden Carrot, the only carrot juice bar in Berkeley. The Golden Carrot is a valuable franchise. Even though you are serving carrot juice in Berkeley, you have student loans to repay so your aim is to maximize profits. The daily demand for carrot juice at the Golden Carrot is $q = 1000 - 50p$ where q is the number of drinks served each day and p is the price per drink. The total daily cost of operating the Golden Carrot is $C(q) = 1000 + 10q$ for $q > 0$.

- (a) What is the Golden Carrot's marginal revenue function?
- (b) What is the Golden Carrot's profit maximizing price? What are the daily profits?

As it turns out upon further scrutiny, the demand described above is the historic demand from students, so we will now write that as $q_S = 1000 - 50p_S$. Now suppose that the Golden Carrot also has demand from tourists, given by $q_T = 1500 - 50p_T$. You can prevent arbitrage and sustain different prices for students and for tourists. In particular, you can set a "Standard Price" and then offer a "Cal Discount" off that price to students who show a Cal ID.

- (c) What standard price should you set? How large should the "Cal Discount" be? What effect, if any, has the appearance of the tourists had on the price paid by students?
- (d) Suppose that you face a rigid capacity constraint at 550 drinks per day due to the limited size of your shop. Your marginal cost is still a constant \$10 per drink up to that capacity. Repeat part (c). Hint: Rather than using the rule $MR_1 = MC$ and $MR_2 = MC$, which works in part (c), you need to use the rule $MR_1 = MR_2$ together with the capacity constraint, $Q_S + Q_T = 550$.